Defeng Zhou

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Education

University of Miami, Coral Gables, FL, USA — Sep. 2025 – Present
Ph.D. in Electrical and Computer Engineering — **Major-course GPA: 3.9/4.0**
Advisor: Prof. Mingzhe Chen, Director of WINS Lab

Sun Yat-sen University, Shenzhen, China — Sep. 2021 – Jun. 2025
B.Eng., School of Intelligent Systems Engineering — **Major-course GPA: 3.8/4.0**
Advisor: Prof. Shimin Gong; top 5% of the cohort in 2021–2022

Research Interests

Efficient multimodal models and spatial reasoning for autonomous driving; semantic communications; reinforcement learning

Selected Publications

1. **D. Zhou**, D. Wei, S. Li, and M. Chen, "Optimization of Collaborative Semantic Communication Network Performance with Channel and Content Preference Feedback," *IEEE Transactions on Wireless Communications*, accepted for publication, 2026.
2. **D. Zhou**, D. Wei, Y. Hu, S. Li, and M. Chen, "Semantic Communication Performance Optimization with Channel and Content Preference Feedbacks," in *Proc. IEEE International Conference on Communications (ICC)*, pp. 1–6, 2026.
3. **D. Zhou**, L. Li, S. Gong, B. Gu, G. Chen, and D. Niyato, "Learning Adaptive Jamming and Beamforming for Hybrid IRS-Assisted Secure NOMA Transmissions," *IEEE Transactions on Communications*, vol. 73, no. 11, pp. 12235–12247, 2025.
4. **D. Zhou**, S. Gong, L. Li, B. Gu, and M. Guizani, "Deep Reinforcement Learning for IRS-Assisted Secure NOMA Transmissions Against Eavesdroppers," in *Proc. International Wireless Communications and Mobile Computing (IWCMC)*, pp. 1236–1241, 2024.

Research Experience

WINS Lab, University of Miami — Aug. 2025 – Present
Research Assistant Advisor: Mingzhe Chen

- **Efficient driving models:** Developing efficient multimodal perception and spatial-reasoning methods for multi-view autonomous-driving systems.
- **Feedback-aware semantic communications:** Designed object-level semantic image transmission with joint subchannel scheduling, power control, and CSI/content-preference feedback; developed critic-guided actor–critic action search, achieving up to $4.16\times$ faster convergence and 66.4% lower weighted MSE than the respective baselines.

Research Experience

WINS Lab, Sun Yat-sen University Jul. 2023 – Apr. 2025

Research Intern Advisor: Shimin Gong

- **Secure uplink NOMA:** First-authored a framework that jointly schedules users and optimizes IRS/AP beamforming; designed a PPO outer loop with AO/SDR inner optimization to exploit multiuser interference against eavesdroppers without dedicated jammers or artificial noise.
- **Hybrid IRS and adaptive jamming:** Developed an energy-constrained active/passive IRS and introduced a dual-cycling PPO–AO framework with a lightweight two-phase approximation that removes nested AO iterations while retaining near-optimal performance, achieving a 54.85% secrecy-rate gain over traditional DRL.

YC Lab, Sun Yat-sen University Sep. 2022 – Nov. 2023

Research Intern Advisor: Calvin Yu-Chian Chen

- **Image deraining:** Implemented PyTorch experiments, including dataset/model setup, baseline reruns, and PSNR/SSIM evaluation; produced qualitative visualizations for a progressive Transformer framework published in *Neurocomputing* (2024).
- **Speaker identification:** Developed a Wav2Vec-based voiceprint-recognition pipeline and evaluated multi-class speaker identification across public and collected datasets.

Teaching

Graduate Teaching Assistant, Digital Design Laboratory, University of Miami

Honors and Awards

Interdisciplinary Talent Award, one of six recipients college-wide, Sun Yat-sen University 2023
Silver Prize, 13th MathorCup Mathematical Modeling Challenge 2023
First Prize, 12th Asia and Pacific Mathematical Contest in Modeling; Best Programming and Best New Media Awards 2022
First Prize Scholarship (top 5%), Sun Yat-sen University 2022

Skills

- Programming and tools: Python, C/C++, MATLAB, R, Git, Linux, LaTeX
- Machine learning: PyTorch, scikit-learn, Gym; vision-language models, computer vision, reinforcement learning

Academic Service

- TPC Member: IEEE International Conference on Communications (ICC); IEEE Vehicular Technology Conference (VTC).
- Reviewer: IEEE JSAC, TMC, TCOM, TGCN, TCCN, and Wireless Communications Letters; *Neural Computing and Applications*; IEEE INFOCOM, ICC, and WF-IoT.